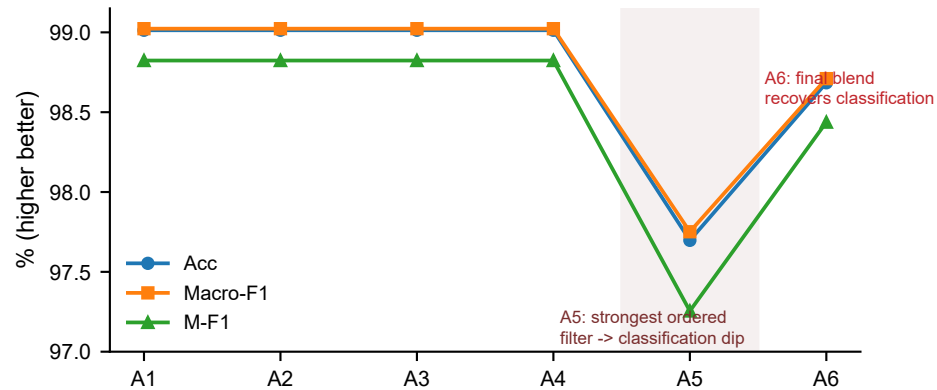
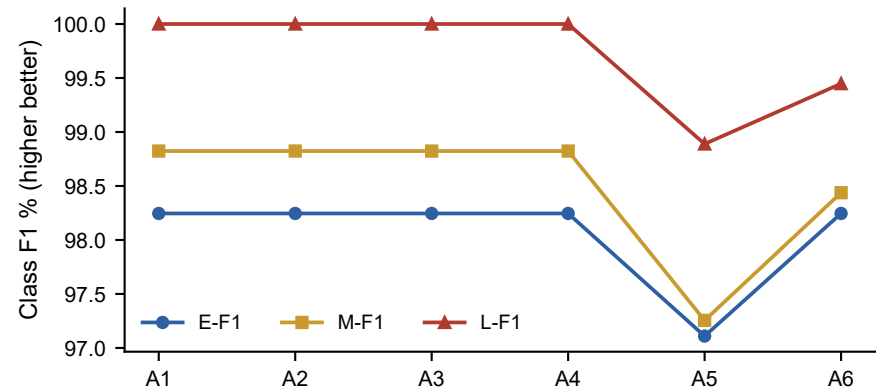


A1→A6 module-progression ablation: predictive performance, middle-stage/transition consistency, and lifecycle probability dynamics (PHM2010 D1, C6 test set, n=304)

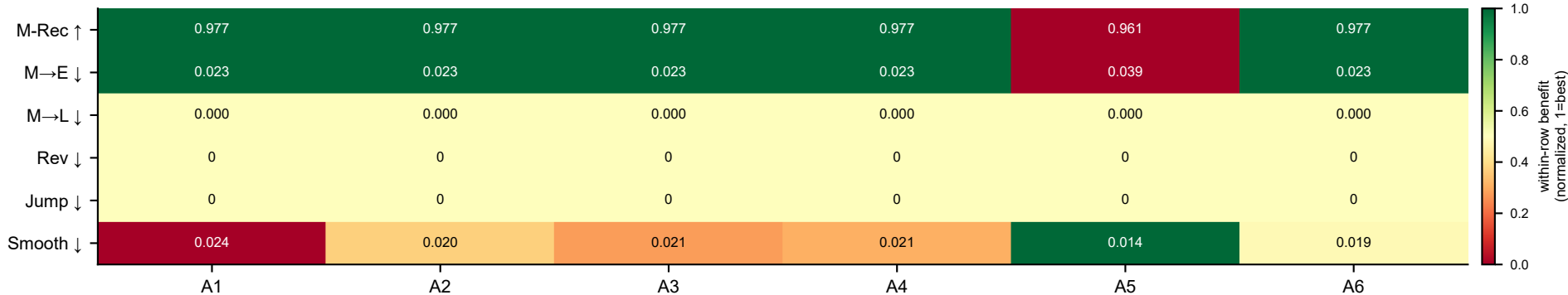
(a) Predictive performance



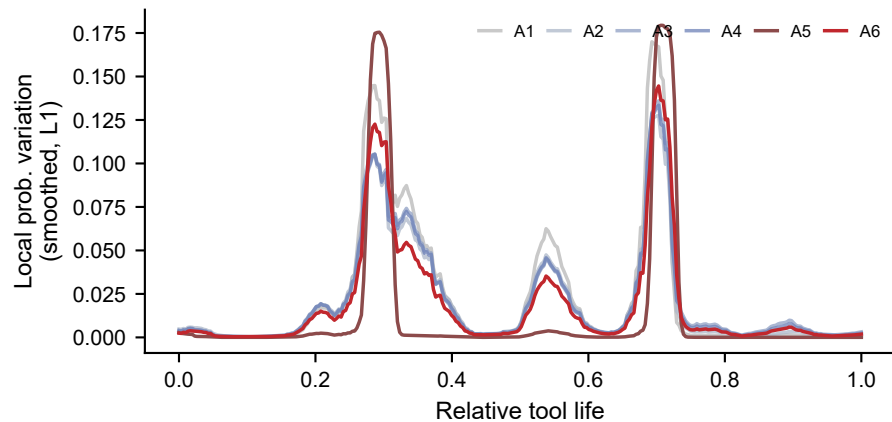
(b) State-wise recognition profile



(c) Middle-stage & transition consistency (color = within-row normalized benefit; numbers = raw values)



(d) Lifecycle local probability variation, A1→A6



(e) Cumulative probability variation, A1→A6

